HARD · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

Circles

01

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ANSWER KEY · SAT QUESTION BANK

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Q1

D

D. 1,768

Choice D is correct. It's given that the radius of the circle is 168 millimeters. Since points M and N both lie on the circle, segments GM and GN are both radii. Therefore, segments GM and GN each have length 168 millimeters. Two segments that are tangent to a circle and have a common exterior endpoint have equal length. Therefore, segment MH and segment NH have equal length. Let x represent the length of segment MH. Then x also represents the length of segment NH. It's given that the perimeter of quadrilateral $GMHN$ is $3{,}856$ millimeters. Since the perimeter of a quadrilateral is equal to the sum of the lengths of the sides of the quadrilateral, $3{,}856 = 168 + 168 + x + x$, or $3{,}856 = 336 + 2x$. Subtracting 336 from both sides of this equation yields $3{,}520 = 2x$, and dividing both sides of this equation by 2 yields $1{,}760 = x$. Therefore, the length of segment MH is $1{,}760$ millimeters. A line segment that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. Therefore, segment GM is perpendicular to segment MH. Since perpendicular segments form right angles, angle GMH is a right angle. Therefore, triangle GMH is a right triangle with legs of length $1{,}760$ millimeters and 168 millimeters, and hypotenuse GH. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2 + b^2 = c^2$. Substituting $1{,}760$ for a and 168 for b in this equation yields $1{,}760^2 + 168^2 = c^2$, or $3{,}125{,}824 = c^2$. Taking the square root of both sides of this equation yields $\pm 1{,}768 = c$. Since c represents a length, which must be positive, the value of c is $1{,}768$. Therefore, the length of segment GH is $1{,}768$ millimeters, so the distance between points G and H is $1{,}768$ millimeters.

Choice A is incorrect. This is the distance between points G and M and between points G and N, not the distance between points G and H.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the distance between points M and H and between points N and H, not the distance between points G and H.

Q2

A

A. $\sqrt{206}$

Choice A is correct. It's given that points A and B lie on the circle with center C. Therefore, $\overline{AC}$ and $\overline{BC}$ are both radii of the circle. Since all radii of a circle are congruent, $\overline{AC}$ is congruent to $\overline{BC}$. The length of $\overline{AC}$, or the distance from point A to point C, can be found using the distance formula, which gives the distance between two points, x_1, y_1 and x_2, y_2, as $\sqrt{x_1 - x_2{}^2 + y_1 - y_2{}^2}$. Substituting the given coordinates of point A, $h + 1, k + \sqrt{102}$, for x_1, y_1 and the given coordinates of point C, h, k, for x_2, y_2 in the distance formula yields $\sqrt{h + 1 - h^2 + k + \sqrt{102} - k^2}$, or $\sqrt{1^2 + \sqrt{102}^2}$, which is equivalent to $\sqrt{1 + 102}$, or $\sqrt{103}$. Therefore, the length of $\overline{AC}$ is $\sqrt{103}$ and the length of $\overline{BC}$ is $\sqrt{103}$. It's given that angle ACB is a right angle. Therefore, triangle ACB is a right triangle with legs $\overline{AC}$ and $\overline{BC}$ and hypotenuse $\overline{AB}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2 + b^2 = c^2$. Substituting $\sqrt{103}$ for a and b in this equation yields $\sqrt{103}^2 + \sqrt{103}^2 = c^2$, or $103 + 103 = c^2$, which is equivalent to $206 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{206} = c$. Therefore, the length of $\overline{AB}$ is $\sqrt{206}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the length of $\overline{AB}$ if the length of $\overline{AC}$ were 103, not $\sqrt{103}$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

28

The correct answer is 28. The equation of a circle in the xy-plane can be written as $(x - t)^2 + (y - s)^2 = r^2$, where the center of the circle is (t, s) and the radius of the circle is r. It's given that circle A is defined by the equation $x^2 + (y - 6)^2 = 7$, which can be written as $(x - 0)^2 + (y - 6)^2 = \sqrt{7}^2$. It follows that $r = \sqrt{7}$ and the radius of circle A is $\sqrt{7}$. It's also given that circle B has the same radius as circle A. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, then $a = r^2$. Substituting $\sqrt{7}$ for r in this equation yields $a = \sqrt{7}^2$, or $a = 7$. It follows that the value of $4a$ is $4(7)$, or 28.

Q4

A

A. 0

Choice A is correct. The sine of a number t is the y-coordinate of the point arrived at by traveling a distance of t units counterclockwise around the unit circle from the starting point $1, 0$. Since the unit circle has a circumference of 2π units, it follows that one full rotation around the circle is equal to a distance of 2π units. Therefore, a distance of 42π units around the circle from the starting point $1, 0$ would result in exactly 21 full rotations, arriving back at the point $1, 0$. So, $\sin 42\pi$ is equal to the y-coordinate of the point $1, 0$, which is 0.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $\cos 42\pi$, not $\sin 42\pi$.

Q5

B

B. $(-10, -8)$

Choice B is correct. The standard equation of a circle in the xy-plane is of the form $(x-h)^2+(y-k)^2=r^2$, where (h, k) are the coordinates of the center of the circle and r is the radius. The given equation can be rewritten in standard form by completing the squares. So the sum of the first two terms, x^2+20x, needs a 100 to complete the square, and the sum of the second two terms, y^2+16y, needs a 64 to complete the square. Adding 100 and 64 to both sides of the given equation yields $(x^2+20x+100)+(y^2+16y+64)=-20+100+64$, which is equivalent to $(x+10)^2+(y+8)^2=144$. Therefore, the coordinates of the center of the circle are $(-10, -8)$.

Choices A, C, and D are incorrect and may result from computational errors made when attempting to complete the squares or when identifying the coordinates of the center.

Q6 A

A. $-\sqrt{3}$

Choice A is correct. A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. If a central angle of a unit circle in the xy-plane centered at the origin has its starting side on the positive x-axis and its terminal side intersects the circle at a point x, y, then the value of the tangent of the central angle is equal to the y-coordinate divided by the x-coordinate. There are 2π radians in a circle. Dividing $\frac{92\pi}{3}$ by 2π yields $\frac{92}{6}$, which is equivalent to $15 + \frac{2}{3}$. It follows that on the unit circle centered at the origin in the xy-plane, the angle $\frac{92\pi}{3}$ is the result of 15 revolutions from its starting side on the positive x-axis followed by a rotation through $\frac{2\pi}{3}$ radians. Therefore, the angles $\frac{92\pi}{3}$ and $\frac{2\pi}{3}$ are coterminal angles and $\tan\frac{92\pi}{3}$ is equal to $\tan\frac{2\pi}{3}$. Since $\frac{2\pi}{3}$ is greater than $\frac{\pi}{2}$ and less than π, it follows that the terminal side of the angle is in quadrant II and forms an angle of $\frac{\pi}{3}$, or $60°$, with the negative x-axis. Therefore, the terminal side of the angle intersects the unit circle at the point $-\frac{1}{2}, \frac{\sqrt{3}}{2}$. It follows that the value of $\tan\frac{2\pi}{3}$ is $\frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}}$, which is equivalent to $-\sqrt{3}$. Therefore, the value of $\tan\frac{92\pi}{3}$ is $-\sqrt{3}$.

Choice B is incorrect. This is the value of $\frac{1}{\tan\frac{92\pi}{3}}$, not $\tan\frac{92\pi}{3}$.

Choice C is incorrect. This is the value of $\frac{1}{\tan\frac{\pi}{3}}$, not $\tan\frac{92\pi}{3}$.

Choice D is incorrect. This is the value of $\tan\frac{\pi}{3}$, not $\tan\frac{92\pi}{3}$.

Q7 100

The correct answer is 100. It's given that point O is the center of a circle and the measure of arc RS on the circle is $100°$. It follows that points R and S lie on the circle. Therefore, $\bar{OR}$ and $\bar{OS}$ are radii of the circle. A central angle is an angle formed by two radii of a circle, with its vertex at the center of the circle. Therefore, $\angle ROS$ is a central angle. Because the degree measure of an arc is equal to the measure of its associated central angle, it follows that the measure, in degrees, of $\angle ROS$ is 100.

Q8

46

The correct answer is 46. It's given that O is the center of a circle and that points R and S lie on the circle. Therefore, $\bar{OR}$ and $\bar{OS}$ are radii of the circle. It follows that $OR = OS$. If two sides of a triangle are congruent, then the angles opposite them are congruent. It follows that the angles $\angle RSO$ and $\angle ORS$, which are across from the sides of equal length, are congruent. Let $x°$ represent the measure of $\angle RSO$. It follows that the measure of $\angle ORS$ is also $x°$. It's given that the measure of $\angle ROS$ is $88°$. Because the sum of the measures of the interior angles of a triangle is $180°$, the equation $x° + x° + 88° = 180°$, or $2x + 88 = 180$, can be used to find the measure of $\angle RSO$. Subtracting 88 from both sides of this equation yields $2x = 92$. Dividing both sides of this equation by 2 yields $x = 46$. Therefore, the measure of $\angle RSO$, in degrees, is 46.

Q9

B

B. $(x - 16)^2 + (y - 17)^2 = 49k^2$

Choice B is correct. The equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r. It’s given that this circle has a center at $(16, 17)$ and a radius of $7k$. Substituting 16 for h, 17 for k, and $7k$ for r in $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 16)^2 + (y - 17)^2 = (7k)^2$, or $(x - 16)^2 + (y - 17)^2 = 49k^2$. Therefore, the equation that represents this circle is $(x - 16)^2 + (y - 17)^2 = 49k^2$.

Choice A is incorrect. This equation represents a circle with radius $7\sqrt{k}$, not $7k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{7k}$, not $7k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{7}\,k$, not $7k$.

Q10 D

D. 25π

Choice D is correct. It's given that the circle is a unit circle, which means the circle has a radius of 1. It's also given that point G is the center of the circle and has coordinates $(0,0)$ and that point H lies on the circle and has coordinates $(-1,y)$. Since the radius of the circle is 1, the value of y must be 0, as all other points with an x-coordinate of -1 are a distance greater than 1 away from point G. Since F and H are points on the unit circle centered at G, let side FG be the starting side of the angle and side GH be the terminal side of the angle. Since side FG is on the positive x-axis and side GH is on the negative x-axis, side FG is half of a rotation around the unit circle, or π radians, away from side GH. Therefore, the positive measure of angle FGH, in radians, is equal to π plus an integer multiple of 2π. In other words, the positive measure of angle FGH, in radians, is an odd integer multiple of π. Of the given choices, only 25π is an odd integer multiple of π.

Choice A is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0,-1)$, not $(-1,0)$.

Choice B is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0,1)$, not $(-1,0)$.

Choice C is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(1,0)$, not $(-1,0)$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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